Geometry Unit 6 Part 1 Assessment Guide

General Information	This assessment covers the benchmarks from lessons 1-6. In Unit 1 students explored dilations and in Unit 4 students focused on the rigid motions found in MA.912.GR.2.1, MA.912.GR.2.3, and MA.912.GR.2.5. Unit 6, lessons 1-6 allow the students to spiral their learning while expanding these benchmarks to include dilations. Teachers can use this assessment after lessons 6 or use the questions from this assessment and part 2 to build one assessment for entire unit. The benchmarks in this unit have similarities. For some of the questions on the assessment the question is asking the student to show knowledge on content that is shared between more than one benchmark, as noted. The nuances of the benchmarks can be found in the language of each benchmark. For the student the application of their knowledge may not seem different. The assessment was created so that teachers have information on student weaknesses on the various transformations while also assessing the similarity transformation of dilation. As the content of rigid motions has been spiraled within the unit, the unit assessment can provide teachers with an understanding of students have grasped both types of transformations. Teachers are encouraged to monitor student understanding of the single transformations as well as when there is an incorporation of a sequence of transformations. If students can understand a single transformation but struggle with a sequence of transformations, then the teacher can offer various strategies focused on understanding how to complete the sequence. Such strategies may include applying each transformation to one vertex at a time or cutting out and physically manipulating the figure, marking the vertices after each transformation.
Calculator Information	All questions are calculator neutral. Students may be given the use of a calculator and it will not have influence on their ability to show their knowledge of the content.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1	Benchmark(s)	MA.912.GR.2.1
Complete the statement. Triangle [○ DNS / ○ KRB] is the result of dilating triangle [○ DNS / ○ KRB] using a scale factor of 3 centered at [○ H. / ○ L. / ○ W.]		**Recommended Scoring** **(4 Total Points)**	No partial credit is suggested.
Question	**2**	**Benchmark(s)**	MA.912.GR.2.1 MA.912.GR.2.3
(table below)		**Recommended Scoring** **(24 Total Points)**	Consider 8 points for each transformation, distributed evenly to the written description and the algebraic description. Consider partial credit if a student's written and algebraic match but it contains a minor error related to the transformation.

	Written Description	Algebraic Description
Transformation 1	Translation left 14, up 7	$(x, y) \rightarrow (x - 14, y + 7)$
Transformation 2	Rotation 90° clockwise about the origin	$(x, y) \rightarrow (y, -x)$
Transformation 3	Dilation of 2.5 centered at the origin	$(x, y) \rightarrow (2.5x, 2.5y)$

Question	3	Benchmark(s)	MA.912.GR.2.1 MA.912.GR.2.3

Transformation of quadrilateral $RDWY$	$LDPC$	$TDMH$	$BXNK$	$SVGP$	none
translation down 2	☐	☐	☐	☐	☐
translation right 6	☐	☐	☐	☐	☐
translation right 3 down 4	☐	☐	☐	☐	☐
dilation by a factor of 2 centered at point D	☐	☐	☐	☐	☐
dilation by a factor of $\frac{1}{2}$ centered at the origin	☐	☐	☐	☐	☐
reflection across y −axis	☐	☐	☐	☐	☐
reflection across $x = -3$	☐	☐	☐	☐	☐
rotation 180° about point D	☐	☐	☐	☐	☐
rotation 180° about the origin	☐	☐	☐	☐	☐

Recommended Scoring

(27 Total Points)

Consider 3 points for each correct match.

Question	4a	Benchmark(s)	MA.912.GR.2.3

☐ ΔBRT maps to ΔYNH using $(x,y) \to (-y,x)$ then $(x,y) \to (x-3,y-4)$ then $(x,y) \to (2x,2y)$

☐ ΔBRT maps to ΔYNH using $(x,y) \to (x-7,y-3)$ then $(x,y) \to (x,-y)$ then $(x,y) \to (2.5x,2.5y)$

☐ ΔBRT maps to ΔYNH using $(x,y) \to (-y,-x)$ then $(x,y) \to (5x,5y)$ then $(x,y) \to (x,-y)$

☐ ΔYNH maps to ΔBRT using $(x,y) \to (x+6,y)$ then $(x,y) \to (x,-y)$ then $(x,y) \to (0.5x,0.5y)$

☐ ΔYNH maps to ΔBRT using $(x,y) \to (0.4x,0.4y)$ then $(x,y) \to (x,-y)$ then $(x,y) \to x+7,y+3)$

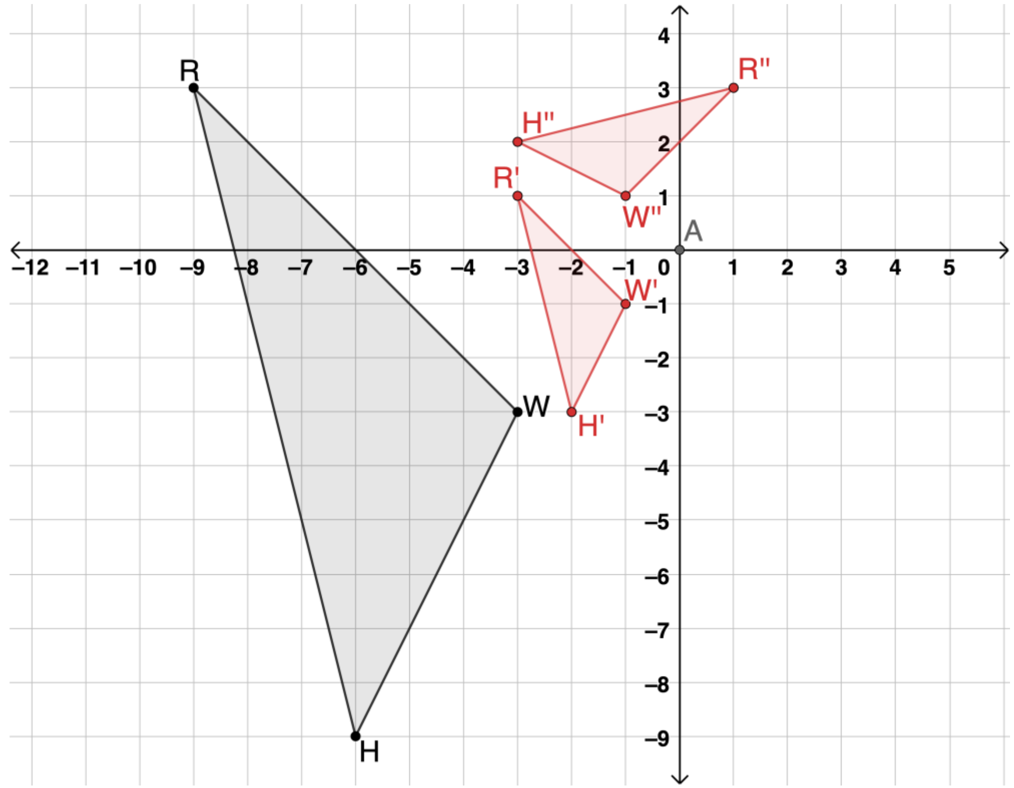
Recommended Scoring

(8 Total Points)

Consider 4 points for each correct selection. Consider deducting 3 points for each additional selection. Consider no points if a student selects all of the choices.

Question	4b	Benchmark(s)	MA.912.GR.2.8
The mapping of triangle BRT to triangle YNH includes a dilation, so therefore the triangles are similar. OR The mapping of triangle YNH to triangle BRT includes a dilation, so therefore the triangles are similar.		**Recommended Scoring** **(5 Total Points)**	Consider 5 points for a correct justification that includes an understanding of dilation.
Question	**5**	**Benchmark(s)**	MA.912.GR.2.1 MA.912.GR.2.3
Reflect ΔRHB across the x-axis, then dilate using a scale of 4.5 centered at the origin.		**Recommended Scoring** **(8 Total Points)**	Consider giving students 4 points for each correct transformation in the sequence.

Question	6	Benchmark(s)	MA.912.GR.2.5



Recommended Scoring

(8 Total Points)

Consider giving students 4 points for the correct dilation and 4 points for the correct rotation. If students do not show the dilation but do have the correct result, then the student should receive full credit. Not all students need to show the mid-step.

Question	7	Benchmark(s)	MA.912.GR.2.2

Sequence of Transformations	Preserves Distance	Preserves Angle Measure	$H''R''N''C''$ is similar but not congruent to $HRNC$
$(x, y) \to (-y, x)$ then $(x, y) \to (3.5x, 3.5y)$	☐	☐	☐
$(x, y) \to \left(\frac{2}{3}x, \frac{2}{3}y\right)$ then $(x, y) \to (x, -y)$	☐	☐	☐
$(x, y) \to (x - 1, y + 5)$ then $(x, y) \to (x, y)$	☐	☐	☐
$(x, y) \to (-x, -y)$ then $(x, y) \to (x + 6, y - 2)$	☐	☐	☐

Recommended Scoring

(16 Total Points)

Consider 4 points for the correct match for each sequence. Partial credit is not suggested.